# UNIFIED THREAT DETECTION AND RESPONSE SYSTEM FOR SMES IN KENYA

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**A PROJECT SUBMITTED TO THE BOARD OF UNDERGRADUATE STUDIES IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER SCIENCE, SCHOOL OF INFORMATION SCIENCE AND TECHNOLOGY, DEPARTMENT OF COMPUTING SCIENCES, KISII UNIVERSITY**

**2024**

# DECLARATION AND RECOMMENDATION

**Declaration by the Candidate (s)**

We declare that this project titled "Unified Threat Detection and Response System for SMEs in Kenya" is our original work and has not been submitted for any degree or examination at any other institution. All information derived from published or unpublished work has been acknowledged in the text.

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## PLAGIARISM DECLARATION

**Declaration by Student (s)**

I/ we declare have read and understood and followed to the letter Kisii University Undergraduate Examination Rules and Regulations.

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# DEDICATION

This work is dedicated to our family and friends for their unwavering support and encouragement throughout our academic journey. Without their guidance and constant backing, this achievement would not have been possible.

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# ABSTRACT

Cybersecurity remains a critical concern for Small and Medium-sized Enterprises (SMEs) globally, particularly in Kenya, where limited resources, low awareness, and evolving threats make these organizations highly vulnerable to cyberattacks. This research presents the development of a **Unified Threat Detection and Response System (UTDRS)** tailored to address the unique cybersecurity challenges faced by SMEs. The system integrates traditional and advanced detection techniques, including signature-based, anomaly-based, and machine learning-driven methods such as neural networks.

The UTDRS employs a modular and scalable architecture, ensuring adaptability to SMEs' financial and technical constraints. Its innovative use of artificial intelligence enhances the detection and mitigation of diverse threats such as phishing, ransomware, Distributed Denial of Service (DDoS), and insider threats. Key advancements over existing systems include a user-friendly dashboard for real-time threat monitoring, automated response mechanisms, and compliance with evolving data protection regulations in Kenya.

By bridging the gap between affordability and advanced functionality, this system empowers SMEs to secure their operations against sophisticated cyberattacks. The research highlights the system's development methodology, including AI model training, architecture design, and validation through simulations of real-world cyber threats. The UTDRS stands as a comprehensive solution, demonstrating the potential to transform cybersecurity for Kenyan SMEs and contributing to the broader discourse on accessible security technologies for small businesses.

# CHAPTER ONE

# INTRODUCTION

## 1.1 Background of the Study

Small and medium-sized enterprises (SMEs) form the backbone of Kenya's economy, contributing significantly to employment and economic growth. However, as SMEs adopt digital tools to enhance operations, they face growing cyber threats that they are often ill-equipped to handle. Cybercriminals increasingly target SMEs due to their limited cybersecurity budgets and lack of advanced security systems, making them easy prey for attacks such as phishing, ransomware, and Distributed Denial of Service (DDoS) attacks (Kshetri, 2017). In Kenya, where SMEs play a critical role in economic development, this vulnerability poses a significant risk to both business continuity and national economic stability (Kithaka, 2020).

Despite the clear risks, many Kenyan SMEs struggle to implement effective cybersecurity measures. Limited resources prevent them from deploying advanced security solutions, leaving them reliant on basic measures such as antivirus software, which is often insufficient in the face of more sophisticated attacks (Mbogo,2018). Moreover, many existing cybersecurity solutions are fragmented and not tailored to the specific needs of SMEs, further exposing them to potential breaches (Scarfone & Mell, 2007).

The rapid digital transformation that many Kenyan SMEs are undergoing has expanded their digital footprint, making them more attractive targets for cybercriminals. Cyberattacks are growing in both number and sophistication, leaving SMEs vulnerable to financial losses, data breaches, and operational disruptions. The evolving threat landscape includes various risks, from malware and phishing to advanced persistent threats (APTs), which SMEs often lack the technical expertise to combat (Kumar et al., 2020). Kenyan SMEs, like their counterparts globally, must address these challenges by adopting more comprehensive and integrated cybersecurity solutions.

Globally, SMEs face similar cybersecurity challenges. The rise of digital transformation has made SMEs worldwide attractive targets for cybercriminals. According to a report by Fortinet, small and medium businesses often sacrifice strong security systems due to budget constraints, making them vulnerable to a wide range of cyber threats. Unified Threat Detection and Response Systems (UTDRS) have emerged as a critical solution, integrating multiple security functions such as network monitoring, endpoint detection, and AI-based threat analysis to provide comprehensive protection. These systems are designed to be cost-effective and scalable, making them suitable for SMEs with limited resources.

In Kenya, the situation mirrors the global landscape but with unique local challenges. The 0 Africa Cybersecurity Report highlights that Kenyan SMEs are particularly vulnerable due to a lack of awareness and inadequate cybersecurity infrastructure. The report emphasizes the need for tailored cybersecurity solutions that address the specific needs of Kenyan SMEs, including affordable and scalable UTDRS. By adopting such systems, Kenyan SMEs can enhance their cybersecurity posture, protect their digital assets, and contribute to national economic stability.

## 1.2 Problem Statement

Kenyan SMEs face significant challenges in implementing robust cybersecurity solutions, leaving them vulnerable to increasingly sophisticated cyber threats. Existing cybersecurity solutions are often fragmented, and many SMEs lack the financial and technical resources to deploy advanced threat detection and response systems. This lack of integrated and scalable systems heightens their exposure to cyberattacks. As a result, there is an urgent need for a Unified Threat Detection and Response System (UTDRS) designed specifically to meet the unique security needs of SMEs in Kenya, providing them with a cost-effective and efficient solution to protect against cyber threats.

## 1.3 Research Objectives

**General Objective**:

To develop a Unified Threat Detection and Response System (UTDRS) that addresses the cybersecurity needs of SMEs in Kenya by integrating real-time threat detection and automated response mechanisms.

**Specific Objectives:**

1. To design and train neural network models optimized for real-time threat identification, focusing on SME-specific cybersecurity challenges such as phishing, ransomware, and insider threats.
2. To develop a unified system architecture that integrates advanced threat detection techniques, including network traffic analysis, endpoint behavior monitoring, and AI-powered anomaly detection.
3. To implement a modular, automated response framework capable of dynamically mitigating diverse cyber threats with minimal human intervention while ensuring adaptability to new attack vectors.

## 1.4 Research Questions

1. What are the key cybersecurity challenges faced by SMEs in Kenya?
2. How can AI, particularly neural networks, enhance real-time threat detection for Kenyan SMEs?
3. What are the critical components of an effective UTDRS for SMEs in Kenya?

## 1.5 Hypothesis

This study hypothesizes that a Unified Threat Detection and Response System (UTDRS) tailored for Kenyan SMEs can significantly enhance their cybersecurity posture by integrating multiple threat detection techniques, utilizing AI for real-time analysis, and automating response processes, thus reducing their exposure to cyberattacks.

## 1.6 Significance of the Study

The study aims to provide Kenyan SMEs with access to advanced cybersecurity solutions by developing a UTDRS specifically tailored to their needs. This unified system will offer SMEs affordable access to sophisticated tools for detecting and mitigating cyber threats, improving their resilience and ability to protect their digital assets. For Kenya's economy, which heavily relies on SMEs, the development and adoption of such a system will strengthen overall cybersecurity and contribute to economic stability. Additionally, the research will provide valuable insights for the broader academic and professional cybersecurity communities, particularly in the context of SMEs in developing economies.

## 1.7 Justification

The necessity for improved cybersecurity measures among Kenyan SMEs is evident given their vulnerability to cyberattacks and the critical role they play in the country's economy. This research seeks to fill the gap in cybersecurity solutions for SMEs by developing a UTDRS that integrates advanced technologies, such as machine learning and AI, making sophisticated cybersecurity accessible to even the most resource-constrained businesses. The system aims to address the unique challenges faced by SMEs, providing a scalable solution that can grow alongside their business needs.

## 1.8 Scope of the Study

This study will focus on designing and evaluating a Unified Threat Detection and Response System specifically for SMEs in Kenya. The research will explore the integration of multiple cybersecurity functions, including real-time threat detection, automated response mechanisms, and AI-driven threat analysis. A prototype will be developed and tested in a simulated environment representative of typical SME operations in Kenya. The scope will also include an evaluation of the system's scalability and applicability to real-world SME environments in Kenya.

## 1.9 Limitations of the Study

The study will be conducted in a simulated environment due to resource constraints, limiting the ability to test the system in live business settings. Furthermore, while the UTDRS will be designed to address a wide range of cyber threats, it may not cover all possible attack vectors. Additional research may be required to expand the system's capabilities and optimize it for broader use cases among SMEs in Kenya.

## 1.10 Key Advancements Made from The Existing Systems

The Unified Threat Detection and Response System (UTDRS) for SMEs in Kenya brings several key improvements over existing cybersecurity solutions, particularly in areas of accessibility, automation, adaptability, and user-centric design.

One of the main advances introduced by UTDRS is its cost-effectiveness and accessibility for SMEs. Traditional enterprise-level cybersecurity systems tend to be prohibitively expensive and complex, often requiring dedicated cybersecurity teams for effective management. The UTDRS, on the other hand, provides a scalable, AI-powered solution specifically tailored to SMEs. By using cloud-based deployment options, the system reduces the need for costly on-site infrastructure, making it accessible to SMEs with limited budgets. This focus on affordability and ease of integration with existing systems allows even small organizations to maintain robust cybersecurity defenses, closing a significant gap that has left many SMEs vulnerable to attacks in the past.

Another notable improvement is the integration of AI and machine learning for automated threat detection and response. Traditional security systems largely depend on manual oversight and rely heavily on signature-based detection, which only identifies known threats. The UTDRS uses advanced machine learning models, including neural networks, to detect both known and unknown threats by analyzing patterns and anomalies in network traffic and user behavior. This combination

of signature-based and anomaly-based detection is particularly effective against zero-day attacks and evolving threats that traditional systems may miss. Furthermore, the system's automated response mechanisms allow it to react to threats in real-time, minimizing the time between detection and response, which is crucial for minimizing the impact of attacks.

The user interface (UI) and user experience (UX) of the UTDRS represent significant improvements over existing systems, which are often complex and require specialized knowledge to operate. The UTDRS is designed to be intuitive, with a dashboard that simplifies the visualization of threat levels, real-time monitoring, and response actions. By offering a clear, user-friendly UI, the system reduces the learning curve and allows non-technical users, such as business owners and small IT teams, to manage and understand their cybersecurity status easily. The dashboard also provides customizable reports and alerts, which can be tailored to the SME's specific needs, ensuring that users are informed of critical threats without being overwhelmed by technical details.

Moreover, the UTDRS offers modular scalability that enables SMEs to add or modify features as their needs evolve. Unlike rigid, traditional cybersecurity systems, the UTDRS can be adapted to different threat landscapes and business requirements through modular upgrades. This adaptability ensures that SMEs can continue to rely on the system as they grow, without needing to replace their cybersecurity infrastructure entirely. For example, as business expands, additional detection modules or response protocols can be integrated to cover a broader range of potential threats, providing long-term value and flexibility that traditional systems typically lack.

Finally, UTDRS places a strong emphasis on compliance and data security, with built-in encryption protocols, role-based access controls, and audit trails for tracking and securing sensitive data. Traditional systems often overlook compliance in favor of functionality, but the UTDRS provides

a balance of both, ensuring that SMEs can protect customer data while adhering to regulatory requirements. This focus on compliance is particularly valuable for SMEs in Kenya, where regulatory frameworks are evolving, and data security is becoming increasingly critical.

In summary, the UTDRS improves existing cybersecurity solutions by making advanced security accessible to SMEs, using AI-driven detection and response, providing an intuitive interface, supporting modular scalability, and ensuring compliance with data protection standards. These advancements position the UTDRS as a comprehensive solution tailored to the unique challenges of SMEs, empowering them to safeguard their operations in an increasingly digital and risk-prone environment.

# CHAPTER TWO
# LITERATURE REVIEW

## 2.1 Introduction

In the digital age, the threat of cyberattacks has become increasingly pervasive, impacting organizations of all sizes. This literature review explores the existing body of research on cybersecurity systems, focusing on the distinct challenges faced by SMEs. Emphasis is placed on how artificial intelligence (AI), machine learning (ML), and neural networks are transforming threat detection mechanisms. Furthermore, the discussion will cover the integration of these technologies into unified cybersecurity systems, which aim to streamline security processes for SMEs in Kenya.

## 2.2 Cybersecurity Challenges in SMEs

Cybersecurity challenges in SMEs differ from larger organizations due to limited financial and technical resources. SMEs often rely on basic security tools, such as antivirus software and firewalls, leaving them vulnerable to sophisticated cyber threats. According to Kim, Chen, and Lee (2019), SMEs struggle to maintain adequate cyber-resilience due to their reliance on outdated security practices. Kenyan literature highlights similar issues, where SMEs face constraints in acquiring advanced cybersecurity technologies (Owino, 2020). Many SMEs also lack the skilled personnel needed to defend against rapidly evolving cyber threats (Mugo, 2021), making them attractive targets for cybercriminals.

## 2.3 Related Work

Existing research indicates that Small and Medium-sized Enterprises (SMEs), especially in developing countries like Kenya, are significantly more vulnerable to cyberattacks compared to larger organizations. This vulnerability stems from factors such as limited financial resources, inadequate technical expertise, and a general lack of awareness about the severity of cybersecurity risks (Odera & Wanyonyi, 2018). Unlike larger corporations with dedicated security teams and

budgets, SMEs often rely on minimal cybersecurity measures, such as basic antivirus software and firewalls, which are insufficient against sophisticated threats (Njenga & Wahome, 2022).

Studies in East Africa highlight how SMEs in the region are frequent targets for cybercriminals, particularly through phishing attacks and malware distribution. These attacks exploit the lack of advanced defenses and the generally lower levels of cybersecurity awareness among SME employees (Mutua, 2021). Cybercriminals often perceive SMEs as low-hanging fruit due to their inadequate defenses, compounded by the reliance on legacy systems and outdated software.

Globally, the challenges facing SMEs are not unique to Kenya. A 2019 study by Kshetri found that 43% of cyberattacks target SMEs, as they are less likely to implement comprehensive security measures. Additionally, small businesses often lack incident response plans, leaving them unprepared for breaches when they occur. Studies by Bhattacharya et al. (2021) further emphasize that SMEs in other regions also struggle with balancing operational costs and investing in cybersecurity infrastructure.

In developed countries, there has been greater adoption of AI-driven cybersecurity tools tailored for SMEs. For example, Ponsard et al. (2018) describe how AI-based systems are being deployed to automate threat detection and response, allowing SMEs to mitigate attacks without requiring large IT teams. However, the high cost of these solutions remains a barrier to adoption in developing economies like Kenya.

Kenyan SMEs face unique challenges that amplify their cybersecurity risks. Njenga and Wahome (2022) reveal that most SMEs in Kenya are unaware of the full extent of potential cyber threats. This lack of awareness often translates into limited budget allocation for cybersecurity initiatives. According to Mugo (2021), only 25% of SMEs in Kenya conduct regular cybersecurity training for their employees, increasing the likelihood of successful social engineering attacks such as phishing.

Further research by Owino (2020) highlights that Kenyan SMEs are particularly vulnerable to ransomware attacks due to weak backup systems and inadequate network segmentation. These

vulnerabilities are compounded by the increasing use of digital platforms, such as mobile money and e-commerce, which expose SMEs to sophisticated cyberattacks.

Recent advancements in AI and neural networks offer hope for improving SME cybersecurity. Chakraborty et al. (2021) propose the use of Feedforward Neural Networks (FNNs) for binary classification of threats, enabling SMEs to quickly identify malicious URLs. Similarly, Convolutional Neural Networks (CNNs) have been utilized to analyze network traffic, detecting anomalies indicative of cyber threats.

In Kenya, initiatives such as the Africa Cybersecurity Innovation Hub aim to make AI-based solutions more accessible to SMEs. While these efforts are promising, a study by Mutiso and Kamau (2022) underscores the need for increased government support and incentives to encourage SME adoption of advanced cybersecurity technologies.

### 2.4 Challenges of Existing Techniques

Traditional cybersecurity techniques face significant challenges, especially when it comes to SMEs. Signature-based detection, though effective for known threats, falls short in detecting novel attacks, making SMEs particularly vulnerable to zero-day exploits (Stallings & Brown, 2018). Anomaly-based detection, while capable of identifying previously unknown threats, often generates high false positive rates, leading to alert fatigue among SME security teams. Additionally, AI-driven solutions, while adaptive, require significant computational resources and updates, posing challenges for SMEs with limited budgets (Sommer & Paxson, 2019). Kenyan SMEs are further hindered by a lack of affordable AI tools tailored to their unique needs (Oduor, 2022). The following table summarizes various existing methods for addressing cybersecurity challenges, including their strengths and weaknesses:

Table 1:Various existing methods, their strengths, and weaknesses in addressing cybersecurity challenges.

| Method | Strengths | Weaknesses | References |
|---|---|---|---|
| **Signature-based Detection** | Highly effective for known threats; easy to implement. | Cannot detect unknown (zero-day) threats; requires frequent updates. | Stallings & Brown (2018); Ochieng (2021) |
| **Anomaly-based Detection** | Can detect unknown threats by identifying abnormal behavior. | High false positive rates; requires constant monitoring. | Sommer & Paxson (2019); Njenga & Wahome (2022) |
| **AI-based Detection** | Capable of detecting sophisticated attacks through pattern recognition; can evolve with new threats. | High computational cost; complex to implement and maintain, especially for SMEs. | Bhattacharya et al. (2021); Wachira (2020); Fielder et al. (2018) |
| **Machine Learning (ML)** | Adaptive to new types of threats; reduces human intervention through automation. | Requires large datasets for training; prone to bias and errors in new environments. | Klaper & Hovy (2014); Ponsard et al. (2018) |
| **Neural Networks** | Excellent for detecting subtle or hidden malicious patterns, such as phishing. | High resource demands; difficult for SMEs to adopt due to complexity and cost. | Chakraborty et al. (2021); Mutiso & Kamau (2022); Dhingra & Singh (2022) |
| **Unified Threat Detection (XDR)** | Centralizes security visibility and response; simplifies management for SMEs with limited resources. | Often expensive; requires integration with existing security tools. | Shen & Liang (2019); Mwangi & Mungai (2021); Grimes (2020) |

| | | | |
|---|---|---|---|
| **Traditional Antivirus Software** | Affordable and simple to use; protects against known malware. | Ineffective against sophisticated attacks; limited protection for newer, complex threats. | Odera & Wanyonyi (2018); Mugo (2021) |

## 2.5 Traditional Threat Detection Mechanisms

Traditional threat detection methods are divided into two categories: signature-based and anomaly-based detection. Signature-based detection relies on recognizing known patterns of malicious activity, or "signatures" (Stallings & Brown, 2018). While effective for known threats, this method is limited in its ability to detect new or unknown attacks, which is a significant challenge for SMEs in Kenya, where new forms of cyberattacks are on the rise (Ochieng, 2021). Anomaly-based detection identifies deviations from normal behavior, but its high false positive rates overwhelm the already limited resources of SMEs (Sommer & Paxson, 2019).

## 2.6 Role of Artificial Intelligence in Cybersecurity

AI and machine learning have revolutionized threat detection mechanisms. AI-based systems can quickly analyze large datasets and detect patterns that suggest malicious activity. Bhattacharya et al. (2021) emphasize that machine learning algorithms can adapt to new cyber threats, making them invaluable for organizations. However, AI systems often require substantial computational power and continuous updates, making them difficult for SMEs to adopt. In Kenya, efforts to integrate AI into cybersecurity have begun, but high costs and technical complexity remain significant barriers for SMEs (Wachira, 2020). AI’s potential in automating threat detection can help reduce the burden on SME security teams by improving accuracy and speed (Fielder et al., 2018).

## 2.7 Unified Cybersecurity Systems

Unified cybersecurity systems aim to streamline security tools into a single platform, reducing complexity and enhancing threat detection for SMEs. Shen and Liang (2019) note that such systems are particularly useful for SMEs, offering comprehensive protection without requiring multiple costly tools. Kenyan SMEs stand to benefit significantly from unified systems, as they are often constrained by both budgets and expertise (Mwangi & Mungai, 2021). The adoption of Extended Detection and Response (XDR) systems, which centralize visibility and response capabilities, has gained traction globally but remains limited in Kenya due to resource constraints (Grimes, 2020).

## 2.8 Neural Networks for Threat Detection

Neural networks have demonstrated immense potential in detecting complex cyber threats. Chakraborty et al. (2021) highlight that neural networks can identify subtle indicators of cyberattacks, such as phishing and obfuscated malware. In Kenya, neural networks are being explored as a solution to combat the growing number of cyber threats targeting SMEs (Mutiso & Kamau, 2022). Feedforward Neural Networks (FNN), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs) have all shown promise in detecting various forms of cyberattacks, including phishing and advanced persistent threats (Dhingra & Singh, 2022).

## 2.9 Machine Learning and AI in Unified Threat Detection

Machine learning and AI-driven systems provide a new frontier for threat detection. Klaper and Hovy (2014) emphasize their ability to identify previously unknown attack patterns. Ponsard et al. (2018) further argue that AI can automate threat detection and make these systems more accessible to SMEs. However, Kenyan SMEs face challenges in adopting AI due to cost and complexity (Ndirangu, 2020). There remains a significant need for more affordable AI-driven cybersecurity tools that can be easily integrated into SMEs' operations.

## 2.10 Threats and Attacks to be Addressed by the System

The UTDRS will be designed to resolve a variety of cyberattacks that frequently target SMEs. The following section outlines the types of attacks that will be addressed by the system and how it will mitigate them:

- **Phishing Attacks**:
    - **Detection**: Uses machine learning to analyze URLs and email content for phishing patterns.
    - **Resolution**: Blocks access to malicious links and notifies users.
- **Malware and Ransomware**:
    - **Detection**: Employs both signature-based and anomaly detection techniques to identify abnormal file behavior indicative of malware.

    - **Resolution**: Isolates the infected device, preventing further spread, and triggers immediate notification to administrators.
- **Distributed Denial of Service (DDoS) Attacks**:
    - **Detection**: Monitors network traffic for spikes and abnormal patterns characteristic of DDoS attacks.
    - **Resolution**: Throttles traffic filters out malicious requests and redirects resources to critical systems to ensure availability.
- **SQL Injection**:
    - **Detection**: Monitors database query logs for suspicious activity patterns associated with SQL injection attacks.
    - **Resolution**: Blocks harmful queries, logs the event, and alerts the system administrator.
- **Cross-Site Scripting (XSS)**:
    - **Detection**: Analyzes user inputs and HTTP requests to detect malicious script injections.
    - **Resolution**: Sanitizes inputs, preventing execution of the injected scripts.
- **Man-in-the-Middle (MitM) Attacks**:
    - **Detection**: Utilizes encryption and network traffic analysis to detect interception attempts.
    - **Resolution**: Enforces stronger encryption protocols and alerts administrators to suspicious activity.
- **Brute Force Attacks**:
    - **Detection**: Monitors repeated failed login attempts to detect brute force attacks.
    - **Resolution**: Temporarily locks accounts, applies CAPTCHAs, and notifies users of suspicious login activity.
- **Insider Threats**:
    - **Detection**: Tracks user behavior to identify unusual actions, such as accessing sensitive data outside of normal working hours.
    - **Resolution**: Automatically restricts user access, prompts for additional authentication, and logs the event for review.

## 2.11 Summary of Gaps

The literature highlights substantial gaps in the development of cybersecurity systems tailored to SMEs, particularly in Kenya. While neural networks offer promise in advanced threat detection, the complexity and costs associated with these technologies remain barriers for SMEs. There is a pressing need for cost-effective, scalable, and adaptable cybersecurity systems that can meet the specific needs of Kenyan SMEs (Nyaga, 2020). The proposed Unified Threat Detection and Response System (UTDRS) seeks to address these gaps by providing a solution that is both affordable and capable of delivering advanced threat detection capabilities to SMEs.

## 2.12 Summary

The literature demonstrates that while there have been significant advancements in threat detection, there is a need for unified solutions tailored to SMEs, especially in Kenya. AI, particularly neural networks, holds promise for improving threat detection through advanced pattern recognition. The proposed UTDRS aims to address the unique challenges faced by SMEs, providing an affordable, scalable, and effective solution for cybersecurity in Kenya.

# CHAPTER THREE

# METHODOLOGY

## 3.1 Introduction

This chapter will outline the research methodologies that will guide the development of the "Unified Threat Detection and Response System for SMEs." These methodologies will be designed to ensure that the system is scalable, adaptable, and suitable for various SME environments. This approach will involve a combination of quantitative and qualitative research, system design techniques, and neural network algorithms. The chapter will cover research design, data collection methods, system architecture, and neural network models used for threat detection.

## 3.2 Research Design

The study will employ an **applied systems development approach** focused on creating a functional prototype of the UTDRS. This approach will involve constructing the prototype, testing its effectiveness across simulated cybersecurity scenarios, and gathering feedback to enhance its performance iteratively. The research design will include a blend of **experimental design, simulation, and real-world evaluation**, allowing the system to adapt to common security challenges encountered by SMEs.

This study will adopt a **mixed-method approach** that combines qualitative insights from cybersecurity experts with quantitative analysis of system performance. This approach will ensure a comprehensive understanding of the requirements and the design of a solution that is user-friendly, scalable, and responsive in real-time.

## 3.3 Data Used for Training the Models

Data for training the neural network models will come from secondary sources, including academic research, industry reports, and open-source cybersecurity incident databases. Emphasis will be placed on the **Common Vulnerabilities and Exposures (CVE) database** and real-time

threat intelligence feeds. These sources will provide essential data for constructing detection algorithms capable of identifying both known and unknown threats accurately.

### 3.4 System Architecture

The system's architecture will employ a **modular design** to support scalability and flexibility across various cybersecurity environments. This modular structure will allow the system to adapt to the specific requirements of each SME and will consist of the following key components:

- **Data Collection Layer**: This layer will gather security logs and other critical data from network systems, applications, and endpoints through lightweight sensors. It will serve as the foundation for subsequent threat analysis.
- **Threat Detection Layer**: Using both **signature-based and anomaly-based detection mechanisms**, this layer will identify potential threats. This layer will integrate artificial neural networks (ANNs), which will be trained on historical data to recognize patterns of known and emerging threats. The layer will utilize **Feedforward Neural Networks (FNNs)** for binary classifications, **Convolutional Neural Networks (CNNs)** for network anomalies, and **Recurrent Neural Networks (RNNs)** to detect sequential data patterns indicative of insider threats.
- **Response Layer**: Upon detecting a threat, the response layer will trigger predefined response protocols. These responses will be automated or manually initiated based on threat severity, providing SMEs with customizable actions to address threats.
- **Dashboard and Reporting Layer**: This user-friendly layer will offer real-time monitoring and analytics, enabling SMEs to track detected threats, review system logs, and configure detection rules and response mechanisms effectively.

### 3.5 Neural Network Models

The UTDRS's core detection capabilities will rely heavily on neural network models due to their high accuracy in detecting complex and evolving cybersecurity threats. Each model will play a unique role in threat detection:

- **Feedforward Neural Networks (FNNs)**: FNNs will perform binary classifications, such as determining whether incoming traffic is malicious or benign. The FNNs will be trained on network traffic data and phishing email datasets, enhancing their effectiveness in recognizing familiar threats.
- **Convolutional Neural Networks (CNNs)**: CNNs, traditionally used in image processing, will analyze network traffic patterns to identify subtle anomalies that indicate cyberattacks. This capability will be especially useful for detecting obfuscated malware, providing an advanced layer of defense.
- **Recurrent Neural Networks (RNNs)**: RNNs will track sequential data, such as login patterns, enabling the system to detect abnormal user behavior over time. This feature will be particularly valuable for identifying insider threats and detecting credential-based attacks.

### 3.6 System Development Tools and Environment

The system will be developed using a suite of advanced tools and technologies to maximize performance and scalability:

- **Programming Languages**: Python will be the primary language for developing neural network models, with TensorFlow serving as the core machine learning library. JavaScript and Node.js will be employed for integrating system components and designing the user interface.
- **Database**: A NoSQL database, MongoDB, will be used to store threat intelligence and network logs. MongoDB's scalability and flexibility make it ideal for managing the large volumes of data generated in a real-time detection system.
- **Cloud Infrastructure**: The system will be deployed on a cloud platform (e.g., AWS or Google Cloud) to ensure scalability and high availability, allowing SMEs to use it as a cloud-based service with minimal infrastructure requirements.
- **Machine Learning Frameworks**: TensorFlow and Keras will be utilized to build and train the neural network models, offering extensive libraries for various network architectures and tools for model validation and optimization.

## 3.7 Software Development Life Cycle (SDLC) Approach

The **Agile Software Development Life Cycle (SDLC)** methodology will be used to guide the development of the UTDRS. The Agile approach is particularly suitable for cybersecurity systems due to its flexibility, iterative structure, and responsiveness to feedback, all of which are essential when dealing with rapidly evolving cyber threats.

### 3.7.1 Rationale for Choosing Agile

The Agile methodology will allow the UTDRS to adapt quickly to the dynamic nature of cybersecurity. Unlike traditional SDLC models, which follow a linear approach, Agile's iterative structure will enable the team to incorporate feedback from SMEs and cybersecurity experts at each development phase. This iterative nature will ensure that the UTDRS continuously improves and adapts to new threats and user requirements.

### 3.7.2 Phases of the Agile SDLC for UTDRS

The following phases will guide the Agile SDLC approach for developing the UTDRS:

1. **Requirement Gathering and Analysis**: Requirements will be gathered at the start of each sprint, enabling the team to stay responsive to emerging cybersecurity trends. This will ensure that each feature developed addresses the most current needs of SMEs.
2. **Design and Prototyping**: Design and prototyping will occur iteratively, allowing for feedback from users after each sprint. Modular components, such as the data collection and threat detection layers, will be prototyped and refined to ensure alignment with SME needs.
3. **Development**: Development will be carried out in incremental cycles, allowing the team to build, integrate, and test system components individually before moving on to the next. Each module—like network traffic monitoring and neural network-based anomaly detection—will be developed separately and integrated with the existing system.

4. **Testing**: Continuous testing will ensure each module functions as intended and integrates well with the rest of the system. Testing will include cross-validation techniques to ensure the reliability of neural network models.
5. **Deployment and Integration**: Agile allows for incremental deployment, meaning foundational system features will be made available for SME use while advanced functionalities are still in development.
6. **Maintenance and Feedback Integration**: Post-deployment, continuous feedback will be collected from SMEs. The feedback will inform subsequent sprints, ensuring the system remains user-centric, responsive to real-world cybersecurity challenges, and adaptable to evolving threats.

### 3.7.3 Benefits of Using Agile for UTDRS

- **Adaptability**: Agile's flexibility will allow the UTDRS to incorporate new detection methods and response protocols for emerging threats.
- **Continuous Feedback**: The Agile methodology will facilitate user feedback at each stage, ensuring the system is developed with direct input from SMEs.
- **Quality Assurance**: Agile's emphasis on iterative testing will ensure high-quality development with each sprint, minimizing risks of errors in the final system.
- **Scalability**: The modular structure of Agile allows the system to be scaled easily, accommodating growing cybersecurity needs as SMEs expand.

The Agile SDLC approach will ensure that the UTDRS is developed efficiently, with regular testing, user-centered enhancements, and the adaptability needed to remain effective in an ever-evolving cybersecurity landscape.

### 3.8 System Testing and Validation

The system will undergo rigorous testing through simulations designed to replicate real-world cybersecurity scenarios. Data from past attacks contributed by SMEs will be used to test the system's detection and response capabilities, ensuring it is well-suited to the cybersecurity needs of SMEs.

#### 3.8.1 Performance Metrics

The system's performance will be evaluated using the following metrics:

- **Accuracy**: The ratio of correctly identified threats to the total number of threats in the test data.
- **False Positives**: The number of benign activities incorrectly flagged as threats.
- **Response Time**: The time taken by the system to detect and respond to a threat, which is crucial for real-time protection.
- **Scalability**: The system's capacity to maintain performance under increasing data volumes and larger network infrastructures.

#### 3.8.2 Validation Techniques

**Cross-validation techniques** will be used to assess the reliability of neural network models. By dividing the dataset into training and validation sets, the system will undergo **k-fold cross-validation** to minimize overfitting and bias, thereby enhancing the model's generalization ability.

### 3.9 Ethical Considerations

Throughout its development and deployment, the UTDRS will adhere to stringent ethical standards in cybersecurity, prioritizing user privacy and data protection. All data used to train the system will be anonymized, and robust access control measures will be implemented to prevent unauthorized access. The system will comply with ethical guidelines and regulatory requirements to ensure the safe and responsible use of sensitive data.

## 3.10 Conclusion

The methodologies outlined in this chapter provide a comprehensive approach for designing, developing, and validating the Unified Threat Detection and Response System (UTDRS) tailored for SMEs. By leveraging advanced neural networks and machine learning algorithms, the system is intended to address modern cybersecurity challenges with high accuracy and scalability. In the subsequent chapter, the system's implementation and the detailed results from testing and validation will be presented.

# CHAPTER FOUR

# SYSTEM ANALYSIS AND DESIGN

## 4.1 Introduction

This chapter presents the system analysis and design of the Unified Threat Detection and Response System (UTDRS) developed for small and medium-sized enterprises (SMEs) in Kenya. The analysis focuses on assessing the requirements, specifications, and functionalities essential for the effective operation of the system. The design section details the modeling of the system architecture, component interactions, and user interfaces, providing a comprehensive understanding of the system's structure and workflow.

## 4.2 System Requirements Analysis

A thorough requirements analysis was conducted to identify the functional and non-functional requirements of the UTDRS. Engaging stakeholders, including cybersecurity professionals and SME representatives, facilitated the collection of insights necessary for defining system needs.

### 4.2.1 Functional Requirements

The functional requirements describe the operations the system must perform:

- **Real-Time Threat Detection**: The system shall detect threats in real-time through multiple detection methods, including signature-based, anomaly-based, and AI-based mechanisms.
- **Incident Response Automation**: The system shall automate predefined response protocols upon threat detection to mitigate potential damages swiftly.
- **User Management**: The system shall have user roles with varying levels of access, including administrators, security analysts, and general users.
- **Reporting and Analytics**: The system shall generate detailed reports on detected threats, response actions taken, and system performance metrics.

- **Dashboard Visualization**: The system shall provide a user-friendly dashboard for centralized monitoring of threats and system status.

### 4.2.2 Non-Functional Requirements

These requirements detail the system's attributes and quality measurements:

- **Scalability**: The system shall support the growth of an SME, managing increased data volumes without performance degradation.
- **Usability**: The system should have an intuitive user interface that minimizes the learning curve for non-technical users.
- **Performance**: The detection system shall process incoming data and provide alerts within a specified time frame (e.g., within seconds).
- **Reliability**: The system shall maintain high availability and recover quickly from any failure, minimizing downtime.
- **Security**: The system shall implement strict access controls and encryption methods to protect sensitive data.

## 4.3 System Design

The design of the UTDRS is based on a modular architecture that enables flexibility and simplicity for SMEs. This modular design separates concerns into distinct components that work together, making the system easier to implement and maintain.

### 4.3.1 System Architecture

The UTDRS employs a layered architecture to separate data collection, processing, and user interface functionalities. The architecture can be divided into four primary layers:

1. **Data Collection Layer**
    - **Components**: Security sensors, agents on endpoints, and network devices.

- **Function**: Collect and aggregate data, such as network traffic logs, application logs, and endpoint security events, to provide comprehensive visibility into the organization's activities.

2. **Threat Detection Layer**
   - **Components**: Signature-based detection engine, anomaly detection engine, and AI models (FNN, CNN, and RNN).
   - **Function**: Analyze collected data using various detection techniques. The AI models utilize trained datasets for detecting known threats and patterns indicative of unknown threats.
3. **Response Layer**
   - **Components**: Incident response protocols, configuration settings, and execution mechanisms.
   - **Function**: Execute automated response actions when a threat is detected (e.g., quarantining an affected system, blocking malicious IP addresses). This layer also allows for manual intervention when necessary.
4. **User Interface Layer**
   - **Components**: User dashboard and reporting tools.
   - **Function**: Provide an interactive dashboard for users to monitor system status, view alerts, generate reports, and configure settings based on their security policies.

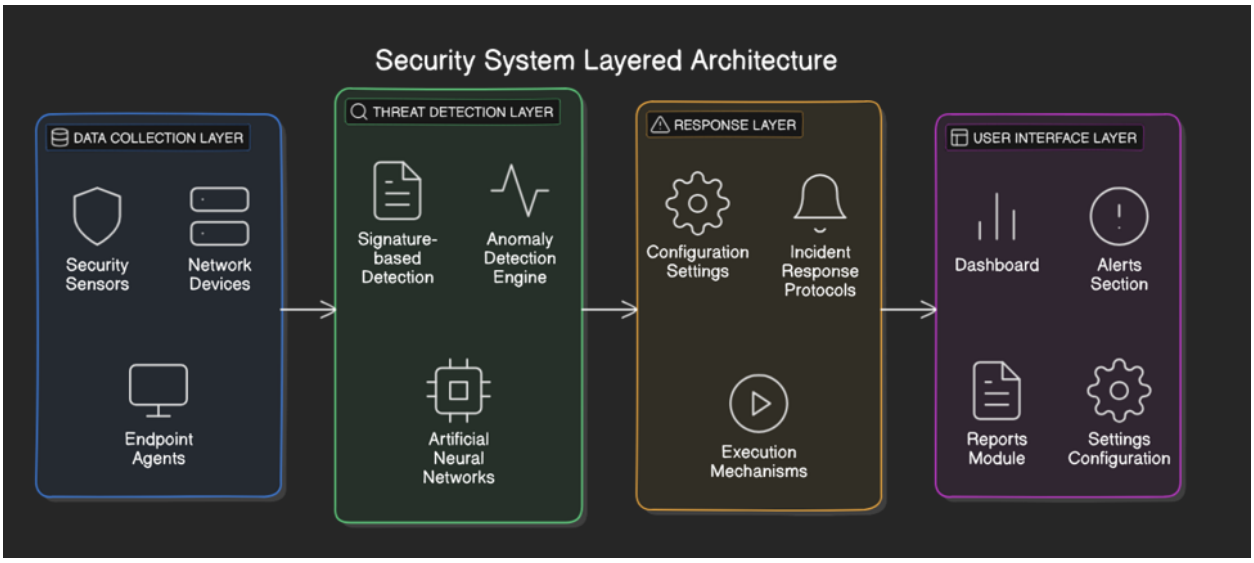



Figure 1:An  illustration of the UTDRS architecture and the interactions between its components.

### 4.3.2 Data Flow Diagram

A Data Flow Diagram (DFD) is created to illustrate how data moves within the system. This diagram represents user interactions, data collection processes, and the flow of information between the layers.

**Figure 2** provides a detailed DFD of the UTDRS, showcasing:

- Data input from sensors and user actions.
- Data processing through threat detection algorithms.
- Output of threat alerts and incident reports.

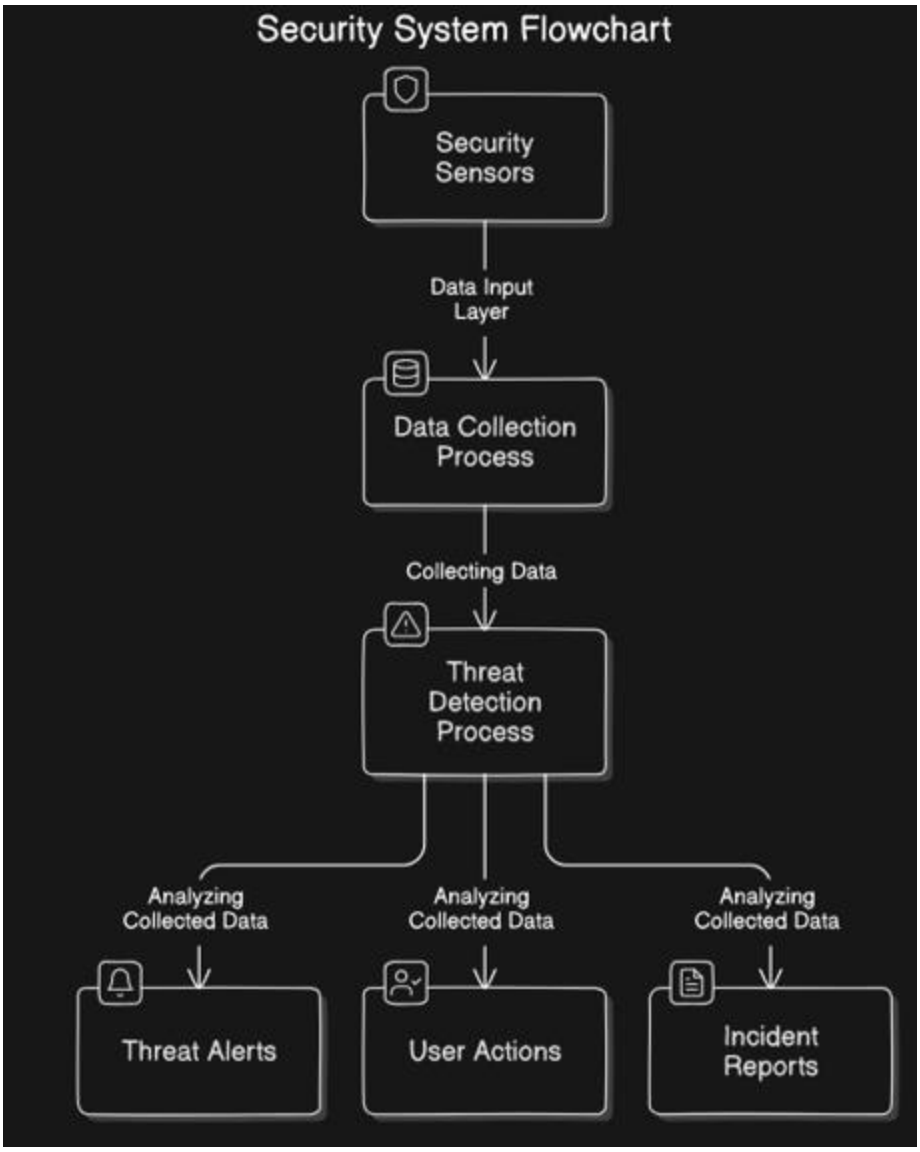



Figure 2: A detailed DFD of the UTDRS.

### 4.3.3 User Interface Design

The user interface is designed to be intuitive and accessible for users with varying technical expertise. The design includes:

- **Dashboard**: Displays real-time alerts, system performance metrics, and threat landscape visualization.
- **Alerts Section**: Shows current and historical alerts, their severity, and current status.
- **Reports Module**: Enables users to generate and view detailed threat analysis reports.
- **Settings Configuration**: Allows users to customize detection parameters, response actions, and user roles.

Prototype mockups of the user interface are provided in **Figure 3**, capturing essential screens and interactions.

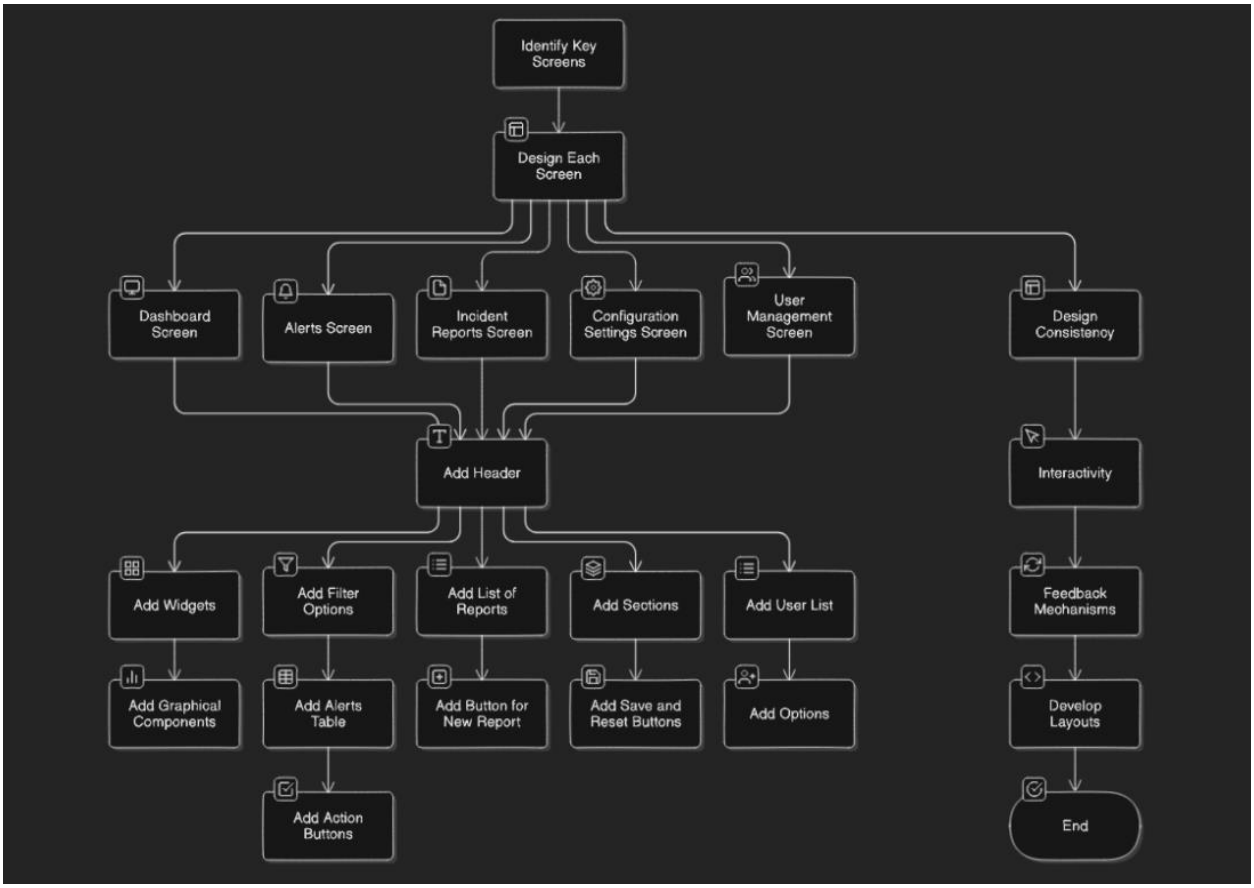



Figure 3: Prototype mockups of the user interface.

## 4.4 Implementation Considerations

When implementing the UTDRS, several considerations were taken into account to ensure successful deployment:

- **Integration with Existing Systems**: The system is designed to integrate seamlessly with existing IT infrastructure and security tools commonly employed by SMEs, minimizing disruption.
- **Training and Support**: Training sessions will be provided for users to familiarize themselves with the system, along with ongoing support to address any queries or concerns.
- **Continuous Improvement**: After deployment, a feedback loop will be established to gather user input for future system enhancements and refinements.

## 4.5 Conclusion

The system analysis and design of the Unified Threat Detection and Response System for SMEs in Kenya have laid a solid foundation for tackling the cybersecurity challenges faced by these enterprises. By addressing both functional and non-functional requirements and employing a modular architecture, the system aims to provide an effective, scalable, and user-friendly solution. The next chapter will detail the system implementation process and present results from the validation and testing phase.

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